Synoptic Question 22: The Physics of a Commercial Airliner

Reading Passage

Read the following article and then answer the questions that follow.

A modern commercial airliner, such as a Boeing 787, is a marvel of applied physics. Its flight can be broken down into several phases: take-off, climb, cruise, and landing. Each phase is governed by a balance of four primary forces: **Lift** (upward), **Weight** (downward), **Thrust** (forward), and **Drag** (backward).

Lift is generated primarily by the wings, which are shaped as aerofoils. This shape forces the air flowing over the top surface to travel faster than the air flowing underneath. According to Bernoulli's principle, this creates a region of lower pressure above the wing, resulting in a net upward force.

Thrust is provided by high-bypass turbofan engines. These engines work by pulling in a large volume of air with a massive fan. A small portion of this air is compressed, mixed with fuel, and ignited in a combustion chamber. This high-energy exhaust is expelled at high speed, but the majority of the thrust (around 80%) comes from accelerating the large volume of "bypassed" air that just passes through the fan. This entire process is a direct application of Newton's third law and the conservation of momentum.

At a cruising altitude of over 10 km, the outside air pressure and temperature are extremely low. The aircraft's cabin must be pressurised to maintain a safe environment for passengers, using compressed air bled from the engines.

Data:

Data for a typical Boeing 787 in operation:

Table 1: Data for Airliner

Parameter	**Value**
Maximum take-off mass, m_{max}	254,000 kg
Total wing area, A_{wing}	360 m^2
Length of take-off runway, d	2800 m
Take-off speed, v_{to}	280 km/h
Cruising speed (constant), v_{cruise}	900 km/h
Cruising altitude, h	12,000 m
Mass flow rate of air (per engine), $\dot{m}$	1100 kg s^{-1}
Exhaust gas velocity (relative to engine), v_{ex}	310 m s^{-1}
Volume of pressurised cabin, V_{cabin}	680 m^3
Pressure in cabin, P_{cabin}	7.50×10^4 Pa (8000 ft equivalent)
Temperature in cabin, T_{cabin}	22.0 °C
Molar mass of air, M_{air}	0.029 kg mol^{-1}
Ideal gas constant, R	8.31 J K^{-1} mol^{-1}

Questions

(a) The aircraft, at its maximum mass, accelerates from rest for take-off.

  (i) Convert the take-off speed of 280 km/h to m s^{-1}.

  (ii) Assuming constant acceleration, calculate the acceleration a required for take-off.

  (iii) Calculate the work done on the aircraft to reach its take-off speed.

(b) The aircraft is now at its cruising altitude and speed.

  (i) Calculate the kinetic energy of the aircraft at its cruising speed.

  (ii) Calculate the gain in gravitational potential energy (GPE) of the aircraft from sea level to its cruising altitude.

(c) While cruising, the aircraft maintains a constant velocity and altitude.

  (i) State the magnitude of the net force acting on the aircraft.

(ii) Calculate the total upward lift force required.

(iii) The lift force is generated by the pressure difference above and below the wings. Using your answer from (c)(ii), calculate the average pressure difference (ΔP) between the top and bottom surfaces of the wings.

(d) The turbofan engines provide the thrust. The thrust F is given by $F = \dot{m}(v_{out} - v_{in})$. Assume the engines are tested on the ground ($v_{in} = 0$) and the air is expelled at v_{ex}.

(i) State the physics principle that explains how the engine generates thrust.

(ii) Calculate the thrust produced by a single engine.

(e) The aircraft cabin is pressurised.

(i) Convert the cabin temperature of 22.0 °C to Kelvin.

(ii) Assuming the air in the cabin behaves as an ideal gas, calculate the total mass of air inside the pressurised cabin.

(f) An observer on the ground hears the sound from the engines as the aircraft passes directly overhead at its cruising speed. The engines emit a sound with a primary frequency of 150 Hz. (Speed of sound at sea level = 340 m s^{-1}).

(i) State the name of the phenomenon that causes the observer to hear a different frequency from the source.

(ii) Calculate the frequency the observer hears as the aircraft moves **away** from them.

Solutions

(a) (i) $v_{to} = 280 \text{ km/h} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = 77.78 \text{ m s}^{-1}$.

(ii) $v^2 = u^2 + 2as \implies a = \frac{v^2 - u^2}{2s}$
$a = \frac{(77.78)^2 - 0}{2(2800)} = 1.079 \text{ m s}^{-2} \approx 1.08 \text{ m s}^{-2}$.

(iii) Work Done $= \Delta KE = \frac{1}{2}mv_{to}^2$
Work Done $= \frac{1}{2}(254000)(77.78)^2 = 7.68 \times 10^8$ J.

(b) (i) $v_{cruise} = 900 \text{ km/h} \times \frac{1000}{3600} = 250 \text{ m s}^{-1}$.
$KE = \frac{1}{2}mv_{cruise}^2 = \frac{1}{2}(254000)(250)^2 = 7.94 \times 10^9$ J.

(ii) $GPE = mgh = (254000)(9.81)(12000) = 2.99 \times 10^{10}$ J.

(c) (i) The velocity and altitude are constant, so acceleration is zero. By Newton's First Law, the net force is **zero**.

(ii) For the vertical forces to be balanced, Lift = Weight.
$Lift = W = mg = (254000)(9.81) = 2.49 \times 10^6$ N.

(iii) $Force = \text{Pressure} \times \text{Area} \implies F_{lift} = \Delta P \times A_{wing}$
$\Delta P = \frac{F_{lift}}{A_{wing}} = \frac{2.49 \times 10^6}{360} = 6917 \text{ Pa} \approx 6920 \text{ Pa}$.

(d) (i) **Conservation of Linear Momentum** (or Newton's Third Law). The engine exerts a force on the air, pushing it backward. By N3L, the air exerts an equal and opposite (forward) force on the engine.

(ii) $F = \dot{m}(v_{out} - v_{in}) = (1100)(310 - 0) = 341000$ N (or 341 kN).

(e) (i) $T(K) = T(°\text{C}) + 273.15 = 22.0 + 273.15 = 295.15$ K.

(ii) Using Ideal Gas Law: $PV = nRT \implies n = \frac{PV}{RT}$
$n = \frac{(7.50 \times 10^4)(680)}{(8.31)(295.15)} = 20800$ mol.
Mass m = moles × Molar mass $= n \times M_{air}$
$m = (20800)(0.029) = 603$ kg.

(f) (i) The **Doppler effect**.

(ii) The source is moving away from the stationary observer.
$f' = f\left(\frac{v_{sound}}{v_{sound} + v_{source}}\right)$
$v_{source} = v_{cruise} = 250 \text{ m s}^{-1}$.
$f' = 150\left(\frac{340}{340+250}\right) = 150\left(\frac{340}{590}\right) = 86.4$ Hz.

Cambridge International AS & A Level Physics (9702)

Paper 4 Structured Questions – Variant 2 (Higher Difficulty)

Context: Commercial Airliner Physics A modern twin-engine airliner (representative values below) is analysed through take-off, climb, cruise and cabin pressurisation.

Given Data

Parameter	Value
Maximum take-off mass, m	$2.54 \times 10^5\,\mathrm{kg}$
Wing area, A	$3.60 \times 10^2\,\mathrm{m}^2$
Take-off runway length, s	$2.800 \times 10^3\,\mathrm{m}$
Take-off speed, v_{to}	$2.80 \times 10^2\,\mathrm{km\,h^{-1}}$
Cruise speed, v_c	$9.00 \times 10^2\,\mathrm{km\,h^{-1}}$
Cruise altitude, h	$1.2000 \times 10^4\,\mathrm{m}$
Mass flow rate per engine, $\dot{m}$	$1.100 \times 10^3\,\mathrm{kg\,s^{-1}}$
Exhaust speed relative to engine, v_e	$3.10 \times 10^2\,\mathrm{m\,s^{-1}}$
Cabin volume, V	$6.80 \times 10^2\,\mathrm{m}^3$
Cabin pressure, P	$7.50 \times 10^4\,\mathrm{Pa}$
Cabin temperature	$2.20 \times 10^1\,°\mathrm{C}$
Molar mass of air, M	$2.9 \times 10^{-2}\,\mathrm{kg\,mol^{-1}}$
Ideal gas constant, R	$8.31\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Speed of sound at sea level	$3.40 \times 10^2\,\mathrm{m\,s^{-1}}$

Assume $g = 9.81\,\mathrm{m\,s^{-2}}$.

Variant 2 Question (9702 P4 style)

(a) Convert the take-off speed and cruise speed to $\mathrm{m\,s^{-1}}$.

(b) The aircraft accelerates uniformly from rest to take-off speed over the runway.

  (i) Determine the acceleration.

  (ii) Determine the time taken for this acceleration.

  (iii) Determine the resultant force acting on the aircraft during this stage.

(c) At cruise, the aircraft is in steady horizontal flight.

  (i) State the resultant force on the aircraft and explain your answer.

  (ii) Calculate the lift force required.

  (iii) Estimate the average pressure difference between the lower and upper wing surfaces.

(d) For one engine, thrust can be modelled by $F = \dot{m}(v_{out} - v_{in})$. For a ground static test, take $v_{in} = 0$.

  (i) Calculate the thrust of one engine.

  (ii) State and explain the physical principle responsible for thrust generation.

(e) The aircraft climbs from sea level to cruise altitude and then cruises at constant speed.

(i) Calculate the increase in gravitational potential energy.

(ii) Calculate the kinetic energy at cruise speed.

(iii) Hence determine the minimum total mechanical energy change from rest at sea level to cruise conditions.

(f) The cabin is pressurised.

(i) Convert cabin temperature to kelvin.

(ii) Use the ideal gas equation to calculate the amount of substance of air in the cabin.

(iii) Calculate the mass of cabin air.

(g) Engine sound has emitted frequency 1.50×10^2 Hz. The aircraft moves directly away from a stationary observer at cruise speed.

(i) Name the phenomenon responsible for the frequency shift.

(ii) Calculate the observed frequency.

(h) **Open-ended conceptual discussion (higher difficulty)**
"A complete account of aircraft flight cannot rely on single-equation models."
Discuss this statement with reference to **at least three** of the following: lift modelling, thrust efficiency, drag mechanisms, energy transfer during climb, cabin pressurisation, and Doppler measurements in real conditions. Your answer should use clear physics reasoning and appropriate technical language.

Sample Solution – Variant 2

(a) $v_{to} = 280 \times \frac{1000}{3600} = 7.78 \times 10^1\,\mathrm{m\,s^{-1}}$ and $v_c = 900 \times \frac{1000}{3600} = 2.50 \times 10^2\,\mathrm{m\,s^{-1}}$.

(b) Using $v^2 = u^2 + 2as$:

  (i) $a = \frac{77.8^2}{2(2800)} = 1.08\,\mathrm{m\,s^{-2}}$.

  (ii) $t = \frac{v-u}{a} = \frac{77.8}{1.08} = 7.22 \times 10^1\,\mathrm{s}$.

  (iii) $F = ma = (2.54 \times 10^5)(1.08) = 2.74 \times 10^5\,\mathrm{N}$.

(c) Steady horizontal cruise:

  (i) Resultant force is zero (constant velocity implies zero acceleration).

  (ii) $L = W = mg = (2.54 \times 10^5)(9.81) = 2.49 \times 10^6\,\mathrm{N}$.

  (iii) $\Delta P = \frac{L}{A} = \frac{2.49\times10^6}{360} = 6.92 \times 10^3\,\mathrm{Pa}$.

(d) (i) $F = \dot{m}(v_{out} - v_{in}) = 1100(310 - 0) = 3.41 \times 10^5\,\mathrm{N}$ per engine.

  (ii) Conservation of momentum / Newton's third law: engine accelerates air backward, air exerts equal and opposite force on engine.

(e) (i) $\Delta E_p = mgh = (2.54 \times 10^5)(9.81)(12000) = 2.99 \times 10^{10}\,\mathrm{J}$.

  (ii) $E_k = \frac{1}{2}mv_c^2 = \frac{1}{2}(2.54 \times 10^5)(250)^2 = 7.94 \times 10^9\,\mathrm{J}$.

  (iii) Minimum total energy change $= \Delta E_p + E_k = 3.78 \times 10^{10}\,\mathrm{J}$.

(f) (i) $T = 22.0 + 273.15 = 2.9515 \times 10^2\,\mathrm{K}$.

  (ii) $n = \frac{PV}{RT} = \frac{(7.50\times10^4)(680)}{(8.31)(295.15)} = 2.08 \times 10^4\,\mathrm{mol}$.

  (iii) $m = nM = (2.08 \times 10^4)(0.029) = 6.03 \times 10^2\,\mathrm{kg}$.

(g) (i) Doppler effect.

  (ii) $f' = f\left(\frac{v}{v+v_s}\right) = 150\left(\frac{340}{340+250}\right) = 8.64 \times 10^1\,\mathrm{Hz}$.

(h) Sample discussion points: Bernoulli-only lift model is incomplete without momentum deflection and circulation; high bypass engines improve propulsive efficiency by reducing wasted exhaust kinetic energy; drag has induced, parasitic and wave components; climb energy partition requires both mgh and kinetic terms; pressurisation introduces structural fatigue and thermodynamic control issues; real Doppler observations include wind, path geometry and non-monochromatic source effects.

Mark Scheme – Variant 2 (9702 P4 style)

Total: 40 marks

(a) [2]

Both speed conversions correct with units. (1+1)

(b) [6]

- (i) Correct kinematics method and $a = 1.08\,\mathrm{m\,s^{-2}}$ [2]
- (ii) Correct time from $v = u + at$ (or equivalent) [2]
- (iii) Correct use of $F = ma$ and value $2.74 \times 10^5\,\mathrm{N}$ [2]

(c) [5]

- (i) Zero resultant force with valid explanation [1]
- (ii) Lift equals weight and correct value [2]
- (iii) Correct pressure-difference calculation [2]

(d) [4]

- (i) Correct thrust value per engine [2]
- (ii) Correct principle and clear explanation [2]

(e) [6]

- (i) Correct GPE increase [2]
- (ii) Correct cruise kinetic energy [2]
- (iii) Correct total minimum mechanical energy change [2]

(f) [5]

- (i) Correct kelvin conversion [1]
- (ii) Correct ideal-gas method and n [2]
- (iii) Correct mass from $m = nM$ [2]

(g) [4]

- (i) Doppler effect identified [1]
- (ii) Correct formula application and numerical frequency [3]

(h) [8] Open-ended physics quality-of-response

Award up to 8 marks using level descriptors.

- **Level 3 (6–8 marks):** Coherent, well-structured argument; accurate and detailed physics across at least three domains; clear links between models and limitations; appropriate terminology used consistently.
- **Level 2 (3–5 marks):** Some correct physics with partial development; at least two relevant domains discussed; reasoning present but uneven depth or limited integration.
- **Level 1 (1–2 marks):** Limited valid points, mostly descriptive; weak physics linkage; fragmented answer.
- **0 marks:** No creditworthy response.

Examiner guidance: accept equivalent valid approaches and numerical answers within appropriate rounding tolerance when method is correct.